Joseph Farkas

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Education

Georgia Institute of Technology — Atlanta, GA
Bachelor of Science in Computer Science, Devices and Intelligence — *Expected May 2027*
Minors in Robotics and German. Concert Orchestra Member — *GPA: 4.00*

- Relevant Coursework: Data Structures and Algorithms, Objects and Design, Robotics and Perception

Professional Experience

Autonomous Systems Lead, Software Mentor — Oct 2021 – Present
North Gwinnett Robotics — *Suwanee, GA*

- Mentoring $\tilde{1}5$ high school students in sensor fusion and control using C++ for 115lb competition robot
- Integrating computer vision localization with on-the-fly A* pathfinding to power fully autonomous scoring
- World Comp. Autonomous Award & Division Champions. **Rated #1 globally for autonomous performance**

Research Assistant — Jan 2026 – Present
Georgia Tech Engineering Space Policy Lab — *Atlanta, GA*

- Modeling the downrange risk of space launches to inform policy decisions on launch safety and risk mitigation
- Scraping FAA NOTAMs to build a dataset of launch attempts; Plotting risk contours using QGIS

Software Intern — Aug 2024 – May 2025
Viasat Inc. — *Duluth, GA*

- Developed C/C++ firmware and Typescript frontend for prototype three-axis satellite antennas
- Built algorithms for analyzing signal quality and atmospheric modeling; integrated CI pipelines for validation

UX Assistant — Aug 2025 – Present
Georgia Tech PACE High Performance Computing Center — *Atlanta, GA*

- Supporting roughly 5,000 researchers across high-performance computing systems with incident triage
- Handling 30+ tickets weekly, resolving scheduler and accounting issues with a 90%+ resolution rate

Activites

Guidance, Navigation and Control Team — Aug 2025 – Present
Propulsive Landers @ Georgia Tech — *Atlanta, GA*

- Applying PID+LQR, nonlinear MPC and lossless convexification to optimize rocket's fuel in powered descent
- Implementing Chebyshev-Gauss-Lobatto nodes for faster on-device trajectory replanning in Rust
- Prototyping with Monoprop UAV (drone); Aiming for 60m hop with 1000N rocket by EOY 2026

RoboCup: Autonomous Software Team — Aug 2025 – Present
RoboJackets @ Georgia Tech — *Atlanta, GA*

- Building multi-agent pathfinding, obstacle avoidance and task allocation in C++ for 6v6 robot soccer

Projects, Presentations & Hackathons

Holonomic Auto-Alignment using CV Localization | *C++, Java, ArUco* — Aug 2023 – May 2025

- Applied second-order kinematics and jerk limiting to replicate S-curve profile for discretized holonomic drive
- Fused CV estimates from fiducials with filtered odometry using a state-space model for real-time robot location
- Allowed competition robot to align to any scoring position with $<$ 1cm accuracy regardless of starting state

Foundations of Autonomous Programming | *Georgia FIRST Robotics Symposium* — Sept 2024

- Presented at statewide symposium on methods for teaching autonomous programming to high school students

soφ: Real-time Adaptive Learning | *Nexhacks (Carnegie Mellon)* | **Devpost** — Jan 2025

- Engineered hallucination-resistant pipeline with RAG + symbolic verification and compression for $<$ 3s latency

Technical Skills

Computer Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML/CSS, Dart, MATLAB, SQL
Frameworks & Libraries: TensorFlow, YOLO, Jupyter, ROS, Matplotlib, OpenCV, ArUco, React
Developer Tools: Git, Linux, Docker, Google Firebase, Android Studio, AWS, Jira
Hardware Experience: Fusion 360, SolidWorks, Autodesk Inventor, 3D Printing, CAN Network, CNC Operation